

Week-5 (Practice Problems)
List

(1)

User will enter a list as a string, your task is to convert the string to a list, if the user enters any number make sure to convert the number to integer.

Note: You are not allowed to use split() function

Input	Output
[7,13,"Hello","Neymar",9.25,"Ronaldo"]	[7,13,"Hello","Neymar",9,"Ronaldo"]
["Hello","Kane","Mabappe",10.88]	["Hello","Kane","Mabappe",10]

(2)

Users will enter a list and a target, your task is to find if there are any two values from the list that can be added together and the target value is achieved. If there is any such case return True otherwise False.

Note: You are not allowed to use split() function

Bonus: Try to do it with 1 loop only

Input	Output
[2,7,3,4,8,11] 9 Explanation: if we add 2 and 7 we can get 9 therefore the output is True	True
[13,4,9,0,8] 2 Explanation: There is no such combination that can be added and 2 can be achieved therefore the output is False	False

(3)

There are two lists, where one contains the name of the students and the other contains marks of each student. Your task is to find the name of the student who got the highest and second highest.

Input	Output
name = ["Ronaldo", "Neymar", "Messi", "Mbappe"] marks = [17,19,9,16]	Highest: Neymar Second Highest: Ronaldo
name = ["Shakib", "Tamim", "Starc", "Morgan"] marks = [2,14,13,9]	Highest: Tamim Second Highest: Starc

(4)

Users will enter a list and your task is to say whether the list is palindrome or not.

Note: You are only allowed to use only one list and you can use split() function

Input	Output
[1,2,3,4,3,2,1]	The list is palindrome
[7,8,8,7]	The list is palindrome
[4,4,4,5,6,4,4,4]	The list is not palindrome

(5)

You are given a list of distance in km, your task is to create a list of distance in m. However the output list should be in reverse order.

Input	Output
[4,13,19,7,12]	[12000,7000,19000,13000,4000]
[0.6,0.085,400,20]	[20000,400000,85.0,600]

(6)

As the election is near, you are given a task to write a code for a digital voting system. You are given a list of markers and their vote count (initially 0). Then there will be another input which will determine how many people can vote. People will choose the marker and based on the input vote count will increase. Finally, show the winner of the election and show the number of votes achieved by each markers.

Input	Output
markers = [{"X":0}, {"Y":0}, {"Z":0}] 5 X X Z Z Z Explanation: X got 2 votes, Y got 0 votes and Z got 3 votes, therefore the winner is Z	Z won the election X got 2 votes Y got 0 votes Z got 3 votes
markers = [{"X":0}, {"Y":0}, {"Z":0}, {"Q":0}] 8 Q Q Z Q X Y X Z	Q won the election X got 2 votes Y got 1 vote Z got 2 vote Q got 3 votes

(7)

Write a python program that will take numbers as input until the user enters "Done". Finally show all the numbers that are prime numbers in a list.

Input	Output
7 4 11 18 Done	[7,11]
19 25 24 13 29 37 40 Done	[19,13,37]

(8)

Messi is worried about his son as he cannot calculate distance between two coordinates, so he requested you to create a calculator for him so that he can pass the exam with ease.

Input	Output
[[4,9],[-2,13]]	7.21
[[11,19],[-7,-19]]	42.04

(9)

Users will always enter a list that contains exactly 5 elements, your task is to find if the sum of the first 4 elements and product of last to element is equal. split() can be used.

Input	Output
[1,3,2,2,4]	Equal Sum: 8 Product: 8
[7,3,2,4,3]	Not Equal Sum: 16

	Product: 12
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(10)

After helping Messi's son Ronaldo requested you to help his son by writing a program that will help his son to find mean, median and mode from the numbers entered. You can assume numbers will always be in sorted order.

Input	Output
2,2,2,4,3,5,5,7	Numbers List: [2,2,2,4,3,5,5,7] Mean: 3.75 Mode: 2 Median: 3.5
2,2,2,4,3,5,5	Numbers List: [2,2,2,4,3,5,5] Mean: 3.28 Mode: 2 Median: 4